Linear Equations & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Solving Multi-Step Linear Equations

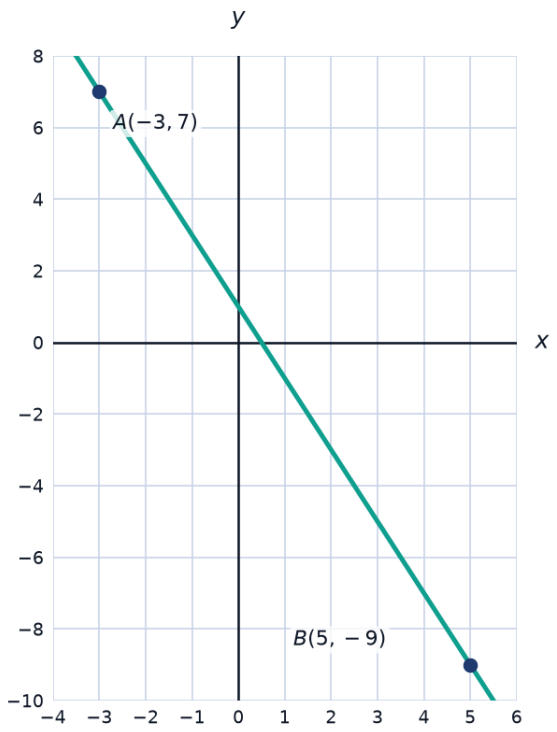
Solve each linear equation for x and verify your solution by substitution:

a $5(2x - 3) - 4 = 3(x + 5)$

b $\frac{3x - 2}{4} + \frac{x + 5}{2} = 7$

2 [Foundational] • [FRQ] • Target: Slope & Intercepts of a Line

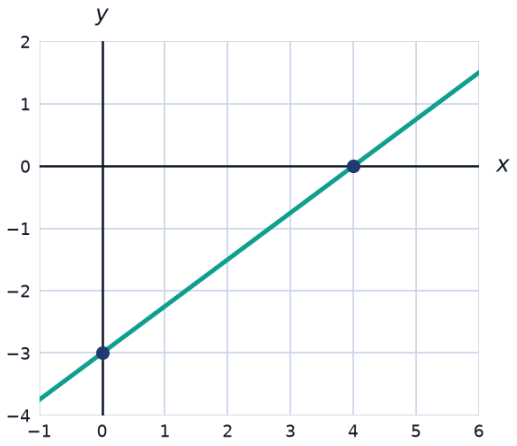
A line passes through the points $A(-3, 7)$ and $B(5, -9)$ in the xy-plane. Determine (a) the slope of the line, (b) its equation in slope-intercept form $y = mx + b$, and (c) its x-intercept.



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3 [Foundational] • [FRQ] • Target: Standard Form & Coordinate Triangle Area

Convert the standard-form linear equation $6x - 8y = 24$ into slope-intercept form, state the slope and both intercepts, and compute the area of the right triangle bounded by the line and the coordinate axes.



4 [Foundational] • [FRQ] • Target: Contextual Linear Modeling

A cloud-computing server plan charges a fixed monthly maintenance fee plus a constant rate per gigabyte of outbound data transfer. Using $120\,\text{GB}$ costs 86 dollars, and using $270\,\text{GB}$ costs 131 dollars. Write a linear function $C(g)$ for the monthly cost (in dollars) of $g\,\text{GB}$, and interpret both the slope and $C(0)$ in context.

5 [Foundational] • [FRQ] • Target: Parallel & Perpendicular Lines

Determine the equation of the line passing through $(-4, 6)$ that is (a) parallel to $3x + 2y = 10$, and (b) perpendicular to $3x + 2y = 10$. Write both answers in slope-intercept form.

6 [Foundational] • [FRQ] • Target: Isolating Quantities in Literal Equations

Rearrange the literal thermal-expansion formula $L = L_0(1 + \alpha(T - T_0))$ to express the temperature T in terms of $L, L_0, \alpha,$ and T_0.

7 [Foundational] • [FRQ] • Target: Conditions for No Solution vs. Infinitely Many Solutions

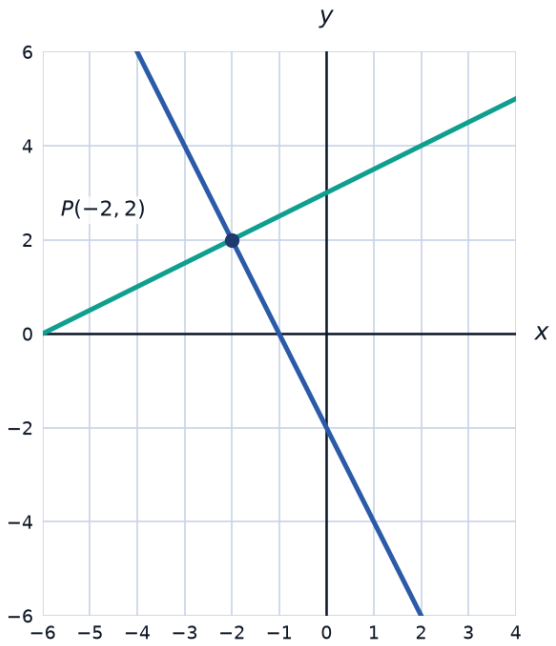
Find the value of the constant k for which the linear equation $4(3x - k) + 6 = 2(6x + 9)$ has infinitely many real solutions.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

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8 [Intermediate] • [FRQ] • Target: Perpendicular Line Intersection

In the xy-plane shown, line ℓ_1 has equation $y = \frac{1}{2}x + 3$ and line ℓ_2 passes through $(0, -2)$ and $(4, -10)$ so that $\ell_1 \perp \ell_2$. Find the exact coordinates of their intersection point P.



9 [Intermediate] • [FRQ] • Target: Linear Mixture & Concentration Modeling

An chemist mixes a 18% saline solution with a 42% saline solution to produce 600 mL of a 28% saline solution. How many milliliters of each solution must be used?

10 [Intermediate] • [FRQ] • Target: Absolute Value Linear Equations

Solve the absolute value equation $3|2x - 5| - 7 = 14$ and plot both solutions on a number line, stating the midpoint and distance from the midpoint to each solution.



11 [Intermediate] • [FRQ] • Target: Parametric Linear Equations with No Solution

For what value(s) of the parameter a does the linear equation $(2a - 5)x + 3a = 7x + 12$ have no solution?

12 [Intermediate] • [FRQ] • Target: Distance-Rate-Time Linear Modeling

A delivery drone flies from depot A to warehouse B against a steady headwind at 36 km/h and returns along the same route with a tailwind at 54 km/h. If the total round-trip flight time is 2.5 hours, find the one-way distance d between A and B and the average speed for the entire round trip.

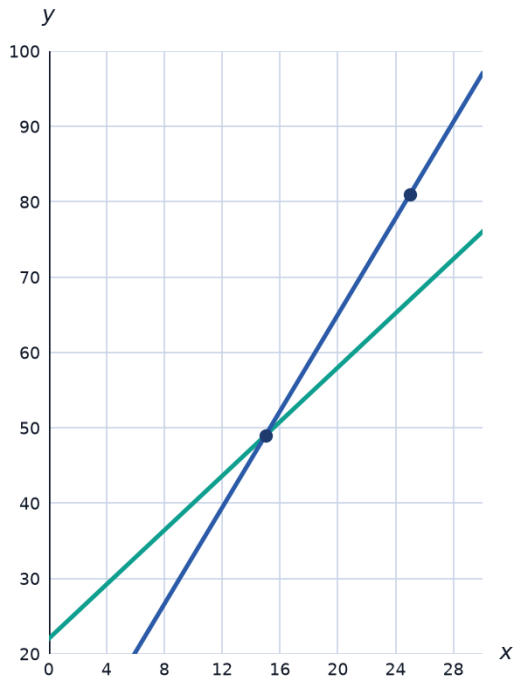
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13 [Intermediate] • [FRQ] • Target: Linear Function Rate of Change & Functional Equations

A linear function f satisfies $f(2k) - f(k) = 6k$ for all real k, and $f(-1) = 4$. Find the formula for $f(x)$ and evaluate $f(10)$.

14 [Intermediate] • [FRQ] • Target: Piecewise Linear Tariff Modeling

A municipal water utility bills residential customers using a two-tier piecewise linear schedule: 22 dollars base fee plus 1.80 dollars per HCF for the first 15 HCF, and 3.20 dollars per HCF for every HCF above 15. If a household's monthly bill was 81.00 dollars, how many HCF of water did the household consume?

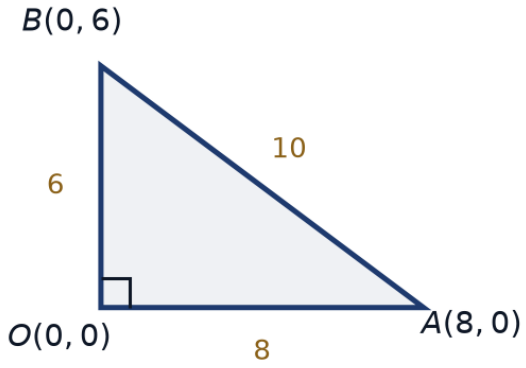


Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • [FRQ] • Target: Perpendicular Projection & Coordinate Geometry • Cross-Lesson: Topic 7.01 (Triangles, Lines & Angles)**

Line ℓ has equation $3x + 4y = 24$ and intersects the x-axis at A and the y-axis at B. Find (a) the length of the altitude h from the origin $O(0, 0)$ to segment AB, and (b) the coordinates of the foot of the perpendicular H on segment AB.



16 **[Challenging] • [FRQ] • Target: Linear Function Composition & Inverse • Cross-Lesson: Topic 2.01 (Linear Systems & Fixed Points)**

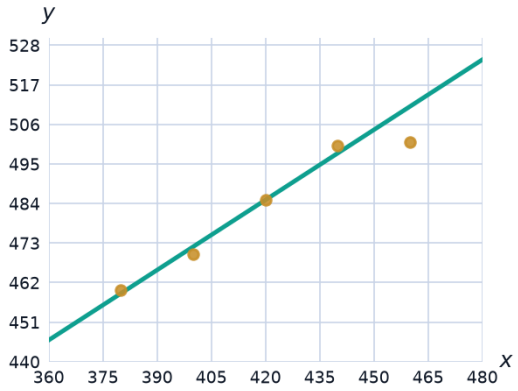
Suppose $f(x) = ax + b$ is a linear function with $a > 0$ such that the composite function satisfies $f(f(x)) = 9x - 16$ for all real x. Find the exact values of a and b, and solve $f(x) = f^{-1}(x)$.

17 **[Challenging] • [FRQ] • Target: Parametric Intercepts & Optimization • Cross-Lesson: Topic 3.01 (Quadratics & Extremum Analysis)**

A line with negative slope m passes through the fixed point $P(3, 4)$ and forms a triangle in the first quadrant with the positive x- and y-axes. Express the x-intercept and y-intercept in terms of m, and verify that if $m = -\frac{4}{3}$, the area of the first-quadrant triangle is minimized at $A = 24$.

18 **[Challenging] • [FRQ] • Target: Linear Regression Centroid & Residual Analysis • Cross-Lesson: Topic 6.01 (Statistics, Scatterplots & Residuals)**

In a bivariate study of $n = 8$ electric vehicles, the mean battery mass is $\bar{x} = 420\,\text{kg}$ and the mean driving range is $\bar{y} = 485\,\text{km}$. The ordinary least-squares regression line is $\hat{y} = 0.65x + b$. (a) Use the centroid property $(\bar{x}, \bar{y})$ to find b. (b) If a vehicle with a $460\,\text{kg}$ battery has an actual range of $501\,\text{km}$, compute its residual $e = y - \hat{y}$.



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19 [Challenging] • [FRQ] • Target: Point-to-Line Distance & Tangent Circle • Cross-Lesson: Topic 9.01 (Circles & Coordinate Geometry)

The line $y = \frac{3}{4}x + 5$ is tangent to a circle centered at $C(4, -2)$. Find the perpendicular distance from $C(4, -2)$ to the line, and write the standard equation of the circle.

20 [Challenging] • [FRQ] • Target: Linear Calibration & Percentage Rate Comparison • Cross-Lesson: Topic 5.01 (Ratios, Rates & Percentages)

In a laboratory calibration, temperature C (in degrees Celsius) and resistance R (in ohms) are linearly related. At $C = 20$, $R = 108.0\,\Omega$; at $C = 95$, $R = 138.0\,\Omega$. Meanwhile, a second sensor has resistance $R_2(C) = 96.0 + 0.65C$. At what temperature C do both sensors have the exact same resistance, and what is the percentage increase in the first sensor's resistance from $C = 0$ to $C = 100$?

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Structure in Linear Equations

If $4(2x - 3) + 5 = 3x + 18$, what is the value of $5x - 7$?

a. 18

b. 23

c. 25

d. 30

22 [Foundational] • [MCQ] • Target: Point-Slope & Standard Form

Which of the following is an equation of the line in the xy-plane that has slope $-\frac{3}{5}$ and passes through $(10, -2)$?

a. $3x + 5y = 20$

b. $3x - 5y = 40$

c. $5x + 3y = 44$

d. $3x + 5y = -20$

23 [Foundational] • [MCQ] • Target: Intercepts from Intercept Form

What is the x-intercept of the graph of $\frac{x}{6} - \frac{y}{4} = 3$ in the xy-plane?

a. $(-12, 0)$

b. $(6, 0)$

c. $(12, 0)$

d. $(18, 0)$

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24 [Foundational] • [MCQ] • Target: Interpreting Slope in Context

A botanist models the height $h(t)$, in centimeters, of a bamboo shoot t days after planting by $h(t) = 4.5t + 12$. Which statement best interprets the number 4.5 in this context?

a. The initial height of the shoot was 4.5 cm.

b. The shoot grows an average of 4.5 cm per day.

c. The shoot reaches 12 cm after 4.5 days.

d. The height increases by 12 cm every 4.5 days.

25 [Foundational] • [MCQ] • Target: Perpendicular Slopes

Line p is defined by $2y - 6x = 9$. Line q is perpendicular to line p in the xy-plane. What is the slope of line q?

a. -3

b. $-\frac{1}{3}$

c. $\frac{1}{3}$

d. 3

26 [Foundational] • [MCQ] • Target: Rearranging Multivariable Formulas

The kinetic energy relation with linear drag work is $E = E_0 - fd$, where E_0 is initial energy, f is friction force, and d is distance. Which equation correctly expresses d in terms of $E, E_0,$ and f?

a. $d = \frac{E - E_0}{f}$

b. $d = \frac{E_0 - E}{f}$

c. $d = f(E_0 - E)$

d. $d = E_0 - \frac{E}{f}$

27 [Foundational] • [MCQ] • Target: Linear Equations with Fractional Coefficients

If $\frac{2x + 5}{3} - \frac{x - 1}{2} = 4$, what is the value of x?

a. 7

b. 11

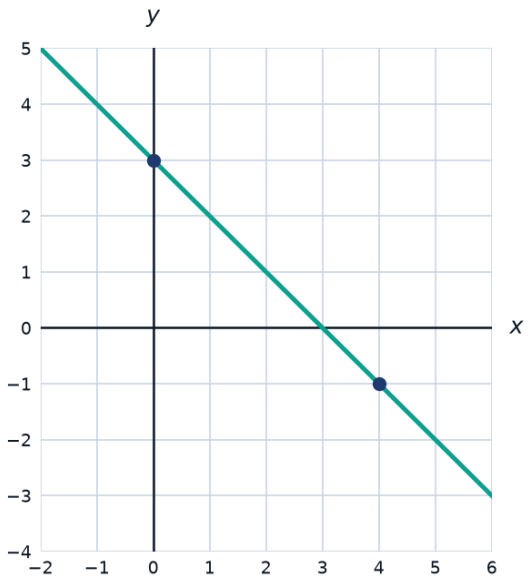
c. 13

d. 17

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28 [Foundational] • [MCQ] • Target: Reading & Extrapolating Linear Graphs

The graph of a linear function f is shown below. What is the value of $f(8)$?



a. -7

b. -5

c. -3

d. -1

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: No-Solution Parameter Conditions

In the equation $k(3x - 4) = 6(2x - 5) + 2$, k is a constant. If the equation has no solution, what is the value of k?

a. 2

b. 3

c. 4

d. 6

30 [Intermediate] • [MCQ] • Target: Piecewise Linear Word Problems

A catering company charges 450 dollars for up to 20 guests, plus 18.50 dollars for each guest beyond 20. If a client's total bill before tax was 968 dollars, how many guests attended the event?

a. 28

b. 44

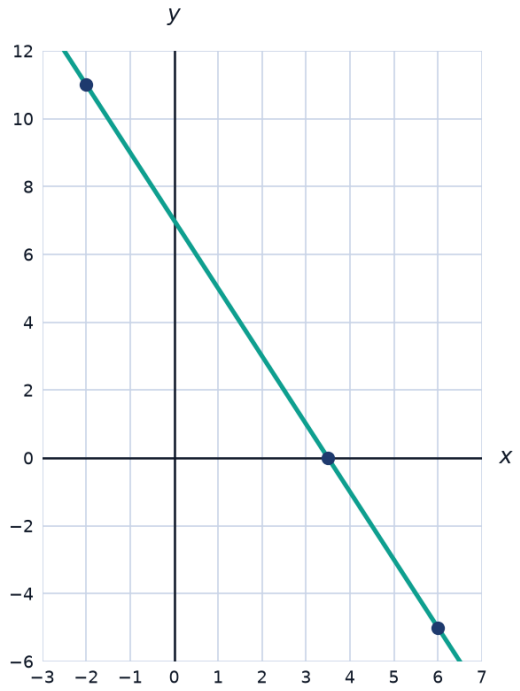
c. 48

d. 52

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31 **[Intermediate]** • **[MCQ]** • **Target: Finding Intercepts from Function Values**

If f is a linear function such that $f(-2) = 11$ and $f(6) = -5$, what is the x-intercept of the graph of $y = f(x)$?



a. $(2.5, 0)$

b. $(3, 0)$

c. $(3.5, 0)$

d. $(4, 0)$

32 **[Intermediate]** • **[MCQ]** • **Target: Absolute Value Non-Negativity**

How many real solutions does the equation $|4x - 9| + 6 = 2$ have?

a. Zero

b. One

c. Two

d. Infinitely many

33 **[Intermediate]** • **[MCQ]** • **Target: Sum of Solutions of Absolute Value Equations**

What is the sum of all solutions to the equation $2|3x - 6| = 18$?

a. 2

b. 4

c. 5

d. 6

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34 [Intermediate] • [MCQ] • Target: Algebraic Slope Coordinates

Line L passes through $(k, 3)$ and $(9, 2k - 1)$ in the xy-plane and has slope $\frac{3}{2}$. What is the value of k?

a. 4

b. 5

c. 6

d. 7

35 [Intermediate] • [MCQ] • Target: Intercept Form & Parametric Slope

The line $\frac{x}{a} + \frac{y}{b} = 1$ passes through $(3, 4)$ and has slope $-\frac{2}{3}$. What is the value of $a + b$?

a. 12

b. 15

c. 16

d. 18

36 [Intermediate] • [MCQ] • Target: Translations of Linear Graphs

When the graph of $y = 4x - 7$ is translated 3 units to the left and 5 units upward in the xy-plane, what is the y-intercept of the resulting line?

a. $(0, 5)$

b. $(0, 7)$

c. $(0, 10)$

d. $(0, 12)$

37 [Intermediate] • [MCQ] • Target: Multi-Step Linear Discount & Tax Models

A store discounts a jacket by 20% off its original price p, and a customer then applies a 15-dollar coupon before paying 8% sales tax on the reduced amount. If the final total is 108.00 dollars, what was the original price p?

a. 135.00 dollars

b. 140.00 dollars

c. 143.75 dollars

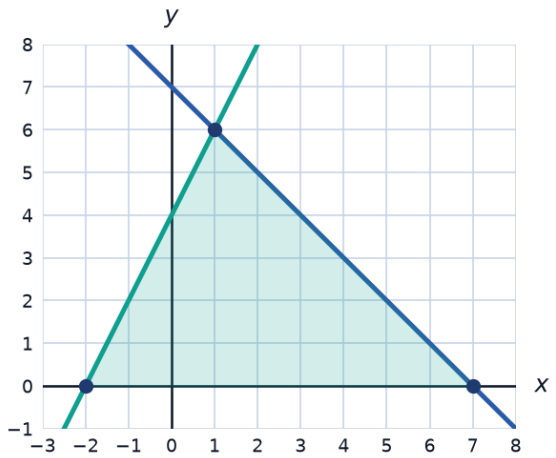
d. 150.00 dollars

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Area Bounded by Intersecting Lines • Cross-Lesson: Topic 2.01 (Linear Systems) & Topic 7.01 (Triangle Area)

What is the area of the triangle enclosed by the lines $y = 2x + 4$, $y = -x + 7$, and the x-axis?



a. 13.5

b. 18

c. 24

d. 27

39 [Challenging] • [MCQ] • Target: Perpendicular Bisector in Coordinate Geometry • Cross-Lesson: Topic 7.01 (Geometry & Midpoint Theorem)

In the xy-plane, the perpendicular bisector of the segment joining $A(-2, 1)$ and $B(6, 5)$ intersects the y-axis at $(0, b)$. What is the value of b?

a. 5

b. 6

c. 7

d. 8

40 [Challenging] • [MCQ] • Target: Linear Functional Equations • Cross-Lesson: Topic 2.01 (Systems of Linear Equations)

For all real numbers x, the linear function g satisfies $2g(x) + g(4 - x) = 6x - 3$. What is the value of $g(5)$?

a. 15

b. 18

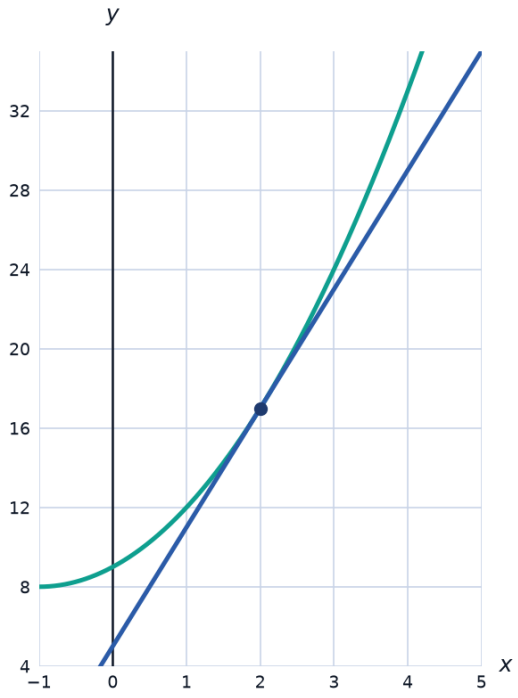
c. 21

d. 24

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41 [Challenging] • [MCQ] • Target: Tangent Line to a Parabola • Cross-Lesson: Topic 3.01 (Quadratics & Discriminant Analysis)

The line $y = mx + 5$ intersects the parabola $y = x^2 + 2x + 9$ at exactly one point in the xy-plane. If $m > 0$, what is the value of m?



a. 4

b. 6

c. 8

d. 10

42 [Challenging] • [MCQ] • Target: Absolute Value Interval Reconstruction • Cross-Lesson: Topic 2.01 (Linear & Absolute Value Inequalities)

If the solution set to the inequality $|3x - a| \leq b$ is the closed interval $[-2, 6]$, what is the value of $a + b$?

a. 12

b. 14

c. 18

d. 20

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43 [Challenging] • [MCQ] • Target: Linear Drainage Rate in a 3D Cylinder • Cross-Lesson: Topic 8.01 (Area, Volume & 3D Solids)

A cylindrical water tank of radius $r = 2\,\text{m}$ and height $H = 10\,\text{m}$ is initially full and drains at a constant volumetric rate so that the water depth $h(t)$ decreases linearly by $0.4\,\text{m}$ per minute. After how many minutes does the tank contain $24\pi\,\text{m}^3$ of water?

a. 8

b. 10

c. 12

d. 15

44 [Challenging] • [MCQ] • Target: Coordinate Reflection Across an Oblique Line • Cross-Lesson: Topic 7.01 (Coordinate Transformations)

The reflection of the point $P(1, 7)$ across the line $y = x + 2$ has coordinates (x_0, y_0). What is the value of $x_0 + y_0$?

a. 6

b. 8

c. 10

d. 12

45 [Challenging] • [MCQ] • Target: Effect of Linear Transformations on Spread • Cross-Lesson: Topic 6.01 (Statistics: Mean & Standard Deviation)

In a data set of 20 observations (x_i, y_i), every point lies on the line $y = -2.5x + 14$. If the standard deviation of the x-values is $s_x = 3.2$, what is the standard deviation s_y of the y-values?

a. -8.0

b. 5.6

c. 8.0

d. 22.0

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